

Exp 6

Code :

```
clear all;
```

```
close all;
```

```
n=input('Enter the number=');
```

```
Hn=1;
```

```
H2n = [Hn Hn; Hn -Hn];
```

```
for i=0:2:n;
```

```
    H2n=[Hn Hn;Hn -Hn];
```

```
    Hn=H2n;
```

```
end
```

```
disp(H2n);
```

```
[p,q]=size(H2n);
```

```
X=input('Enter first waish code number:');
```

```
Y=input('Enter second waish code number:');
```

```
H=H2n(X,1:q).*H2n(Y,1:q);
```

```
disp(H);  
L=0;  
for k=1  
    for z=1:q  
        L=L+H(k,z)  
        z+1;  
    endfor  
endfor  
disp('The Addition is:');  
disp(L);  
if L==0;  
    disp('It is orthogonal');  
else  
    disp('It is non-orthogonal');  
end
```

Output:

Enter the number=2

1 1 1 1

1 -1 1 -1

1 1 -1 -1

1 -1 -1 1

Enter first waish code number:1

Enter second waish code number:2

1 -1 1 -1

L = 1

L = 0

L = 1

L = 0

The Addition is:

0

It is orthogonal